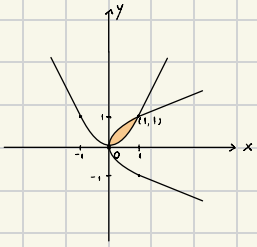
Integrale duble

- Să se reprezinte grafic următoarele domenii:

1. D este domeniul mărginit de parabolele $y = x^2$ și $y^2 = x$



$y = x^2$: $x_v = 0$, $y_v = 0$
Oy axă de simetrie
$x = 1 \Rightarrow y = 1$
$x = -1 \Rightarrow y = 1$

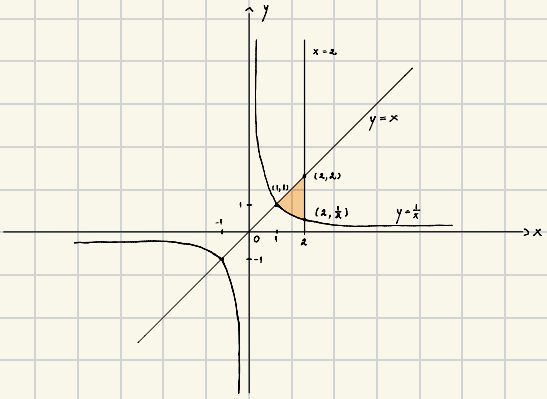
$x = y^2$: $x_v = 0$, $y_v = 0$
Ox axă de simetrie
$y = 1 \Rightarrow x = 1$
$y = -1 \Rightarrow x = 1$

$$\begin{cases} y = x^2 \\ y = \sqrt{x} \end{cases} \Rightarrow x^2 = \sqrt{x} \Rightarrow x^4 = x \Rightarrow x^4 - x = 0 \Rightarrow x(x-1)(x^2+x+1) = 0 \Rightarrow x \in \{0, 1\}$$

$x = 0 \Rightarrow y = 0 \Rightarrow O(0,0)$

$x = 1 \Rightarrow y = 1 \Rightarrow (1,1)$

2. D este domeniul mărginit de dreptele $x = 2$, $y = x$ și hiperbola $xy = 1$



$$\begin{cases} y = \frac{1}{x} \\ y = x \end{cases} \Rightarrow \frac{1}{x} = x \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$x = 1 \Rightarrow y = 1 \Rightarrow (1,1)$

$x = -1 \Rightarrow y = -1 \Rightarrow (-1,-1)$

3. D este domeniul mărginit de curbele $y = 0$, $x + y - 6 = 0$, $y^2 = 8x$

$y = 0 \Rightarrow$ axa Ox

$x + y - 6 = 0 \Rightarrow y = -x + 6$

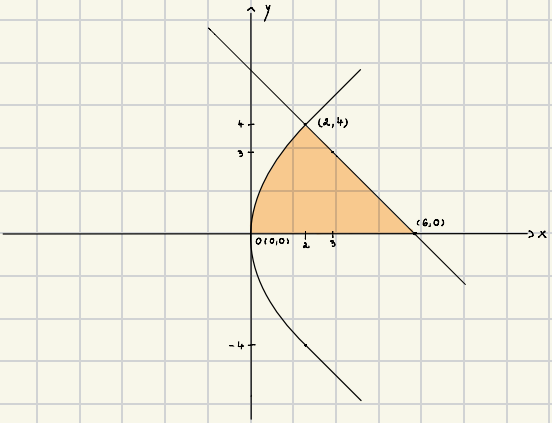
$x = 6 \Rightarrow y = 0$

$x = 3 \Rightarrow y = 3$

$y^2 = 8x \Rightarrow x = \frac{1}{8}y^2 \Rightarrow x_V = 0,\ y_V = 0$

$y = 4 \Rightarrow x = 2$

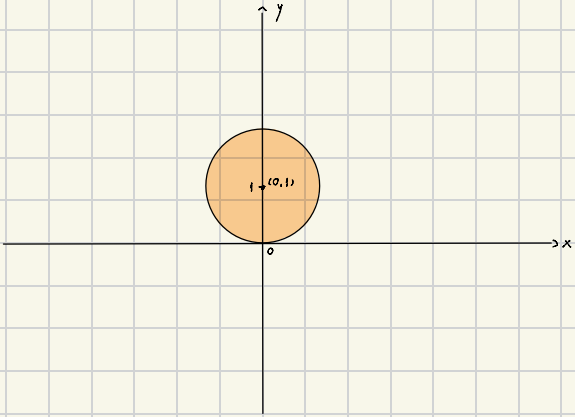
$y = -4 \Rightarrow x = 2$



4. $D = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 2y\}$

$x^2 + y^2 \leq 2y \iff x^2 + y^2 - 2y \leq 0 \iff x^2 + y^2 - 2y + 1 \leq 1 \iff (x-0)^2 + (y-1)^2 \leq 1^2 \Rightarrow$ centru $(0,1)$

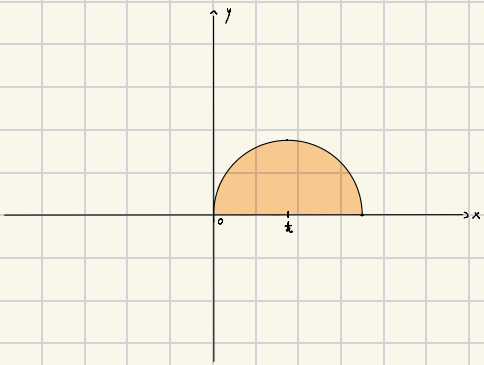
raza $R = 1$



5. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq x,\ y \geq 0\}$

$x^2 + y^2 \leq x \iff x^2 - x + y^2 \leq 0 \iff x^2 - x + \frac{1}{4} + y^2 \leq \frac{1}{4} \iff (x - \frac{1}{2})^2 + (y-0)^2 \leq (\frac{1}{2})^2 \Rightarrow$

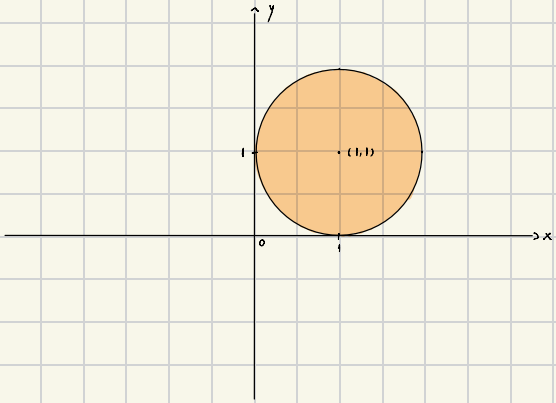
$\Rightarrow$ centrul $(\frac{1}{2}, 0)$ și raza $R = \frac{1}{2}$



6. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 2x + 2y - 1\}$

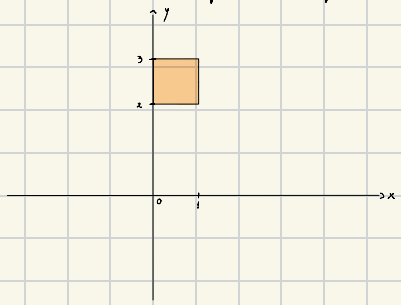
$x^2 + y^2 \leq 2x + 2y - 1 \iff x^2 - 2x + 1 + y^2 - 2y + 1 \leq 1 \iff (x-1)^2 + (y-1)^2 \leq 1^2 \Rightarrow$

$\Rightarrow$ centrul $(1,1)$ și raza $R = 1$



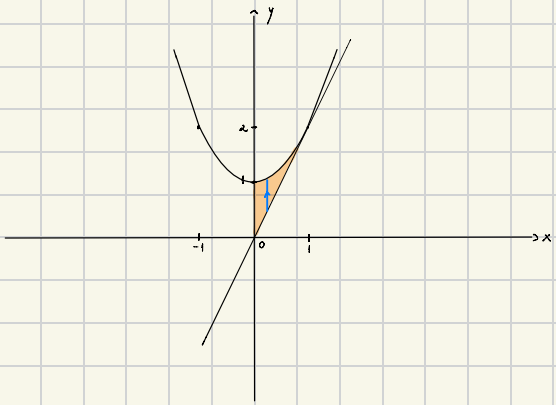
- Să se calculeze următoarele integrale duble $\iint_D f(x,y)\,dx\,dy$, cu reprezentarea grafică a lui D:

7. $D = [0,1] \times [2,3]$, $f(x,y) = xy^2$



$$\iint_D f(x,y)\,dx\,dy = \int_0^1 \left(\int_2^3 xy^2\,dy \right) dx = \int_0^1 \left(x \cdot \frac{y^3}{3} \Big|_2^3 \right) dx = \frac{19}{3} \int_0^1 x\,dx = \frac{19}{3} \cdot \frac{x^2}{2} \Big|_0^1 = \frac{19}{6}$$

8. $D = \{(x,y) \in \mathbb{R}^2 \mid 0 \le x \le 1,\ 2x \le y \le x^2 + 1\}$, $f(x,y) = x$. Calculați și aria (D)



$y = 2x$: $x = 0 \Rightarrow y = 0$; $x = 1 \Rightarrow y = 2$

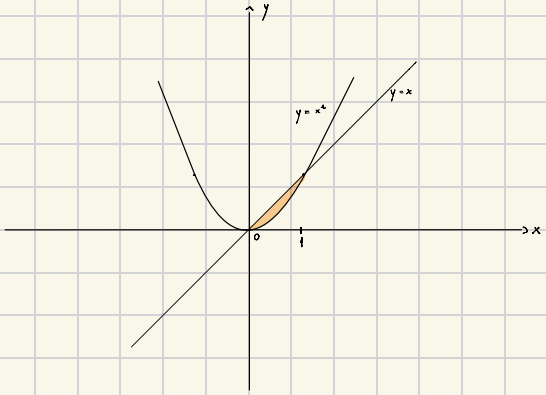
$y = x^2 + 1$: $x_V = 0$, $y_V = 1$; $x = 1 \Rightarrow y = 2$; $x = -1 \Rightarrow y = 2$

$$\iint_D f(x,y)\,dx\,dy = \int_0^1 \left(\int_{2x}^{x^2+1} x\,dy \right) dx = \int_0^1 x(x^2 - 2x + 1)\,dx = \int_0^1 (x^3 - 2x^2 + x)\,dx = \left(\frac{x^4}{4} - \frac{2}{3}x^3 + \frac{x^2}{2} \right) \Big|_0^1 =$$

$$= \frac{1}{4} - \frac{2}{3} + \frac{1}{2} = \frac{3 - 8 + 6}{12} = \frac{1}{12}$$

$$\text{aria}(D) = \iint_D dx\,dy = \int_0^1 (x^2 - 2x + 1)\,dx = \frac{x^3}{3} - x^2 + x \Big|_0^1 = \frac{1}{3} - 1 + 1 = \frac{1}{3}$$

9. D este domeniul mărginit de curbele $y = x$ și $y = x^2$, $f(x,y) = 3x - y + 2$. Calculați și aria(D).



$y = x^2$: $x_V = 0$, $y_V = 0$

$x = x^2 \Rightarrow x(x-1) = 0 \Rightarrow x = 0, x = 1$

$\Downarrow \quad \Downarrow$

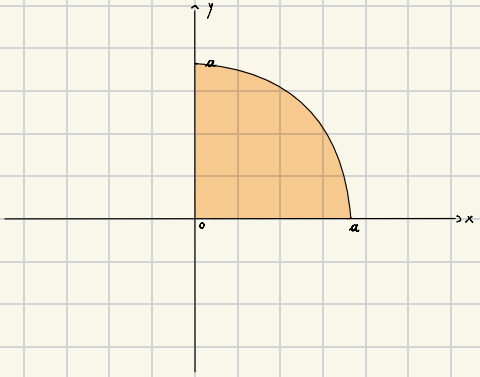
$y = 0 \quad y = 1$

$$\iint_D f(x,y)\,dx\,dy = \int_0^1 \left(\int_{x^2}^{x} (3x - y + 2)\,dy \right) dx = \int_0^1 \left(3x(x - x^2) - \frac{x^2}{2} + \frac{x^4}{2} + 2(x - x^2)\right) dx =$$

$$= \left(x^3 - \frac{3}{4}x^4 - \frac{1}{6}x^3 + \frac{1}{10}x^5 + x^2 - \frac{2}{3}x^3 \right)\Big|_0^1 = 1 - \frac{3}{4} - \frac{1}{6} + 1 - \frac{2}{3} + \frac{1}{10} = \frac{60 - 45 - 10 + 60 - 40 + 6}{60} = \frac{31}{60}$$

$$\text{aria}(D) = \iint_D dx\,dy = \int_0^1 (x - x^2)\,dx = \left(\frac{x^2}{2} - \frac{x^3}{3} \right)\Big|_0^1 = \frac{1}{6}$$

10. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le a^2, a > 0, x \ge 0, y \ge 0\}$, $f(x,y) = \frac{1}{\sqrt{1 + x^2 + y^2}}$



$x = r\cos t$

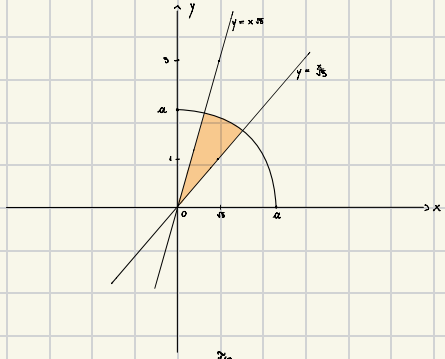
$y = r\sin t$

$dx\,dy = r\,dr\,dt \quad (J(r,t) = r)$

$r \in [0, a]$, $t \in \left[0, \frac{\pi}{2}\right]$

$$\iint_\Delta f(x,y)\,dx\,dy = \int_0^{\pi/2}\left(\int_0^a \frac{r}{\sqrt{1+r^2}}\,dr\right)dt = \int_0^{\pi/2}\left(\sqrt{1+r^2}\,\Big|_0^a\right)dt = \frac{\pi}{2}\left(\sqrt{1+a^2}-1\right)$$

11. $\Delta = \left\{(x,y)\in\mathbb{R}^2 \;\middle|\; x^2+y^2\le a^2,\ 0\le\frac{x}{\sqrt{3}}\le y\le x\sqrt{3}\right\}$, $f(x,y)=x^2+y^2$



$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r\in[0,a]$

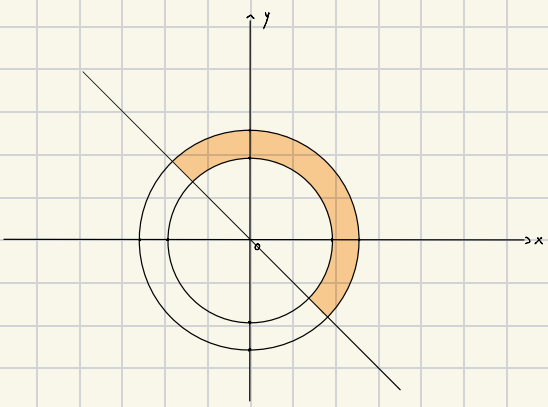
$\frac{r\cos t}{\sqrt{3}} \le r\sin t \le r\cos t\sqrt{3} \Rightarrow$

$\Rightarrow \frac{\cos t}{\sqrt{3}} \le \sin t \le \cos t\sqrt{3} \quad \overset{t\neq\frac{\pi}{2}}{\Rightarrow}$

$\Rightarrow \frac{\sqrt{3}}{3} \le \operatorname{tg} t \le \sqrt{3} \Rightarrow t\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$$\iint_\Delta f(x,y)\,dx\,dy = \int_{\pi/6}^{\pi/3}\left(\int_0^a r^3\,dr\right)dt = \int_{\pi/6}^{\pi/3}\left(\frac{r^4}{4}\Big|_0^a\right)dt = \frac{a^4}{4}\left(\frac{\pi}{3}-\frac{\pi}{6}\right) = \frac{\pi a^4}{24}$$

12. $\Delta = \left\{(x,y)\in\mathbb{R}^2 \;\middle|\; 2\le x^2+y^2\le 4,\ x+y\ge 0\right\}$, $f(x,y)=\sqrt{x^2+y^2}$



$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r\in[\sqrt{2},2]$

$r\cos t + r\sin t \ge 0 \Rightarrow$

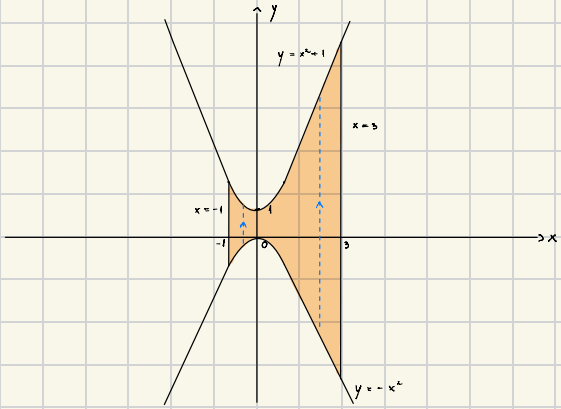
$\Rightarrow \cos t + \sin t \ge 0 \Rightarrow$

$\Rightarrow \sin t \ge -\cos t \quad \overset{t\neq\frac{\pi}{2}}{\Rightarrow}$

$\Rightarrow \operatorname{tg} t \ge -1 \Rightarrow t\in\left[-\frac{\pi}{4},\frac{3\pi}{4}\right]$

$$\iint_\Delta f(x,y)\,dx\,dy = \int_{-\pi/4}^{3\pi/4}\left(\int_{\sqrt{2}}^2 r^2\,dr\right)dt = \int_{-\pi/4}^{3\pi/4}\left(\frac{r^3}{3}\Big|_{\sqrt{2}}^2\right)dt = \frac{8-2\sqrt{2}}{3}\cdot\pi$$

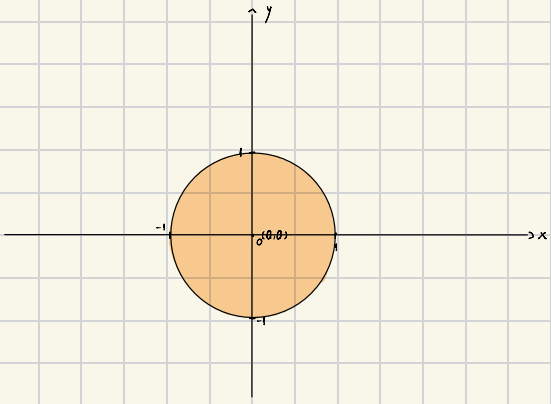
13. D este domeniul mărginit de curbele $y = x^2 + 1$, $y = -x^2$, $x = -1$, $x = 3$, $f(x,y) = x + 3y$



$$\iint_D f(x,y)\,dx\,dy = \int_{-1}^{3}\left(\int_{-x^2}^{x^2+1}(x+3y)\,dy\right)dx = \int_{-1}^{3}\left(x(2x^2+1) + \frac{3}{2}y^2\Big|_{-x^2}^{x^2+1}\right)dx =$$

$$= \int_{-1}^{3}(2x^3 + 3x^2 + x + \tfrac{3}{2})\,dx = \left(\frac{x^4}{2} + x^3 + \frac{x^2}{2} + \frac{3}{2}\right)\Big|_{-1}^{3} = \frac{81}{2} + \frac{54}{2} + \frac{9}{2} + \frac{12}{2} - \frac{1}{2} + \frac{2}{2} - \frac{1}{2} = 78$$

14. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le 1\}$, $f(x,y) = e^{x^2+y^2}$. Calculați și aria (D)



$x = r\cos t$

$y = r\sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

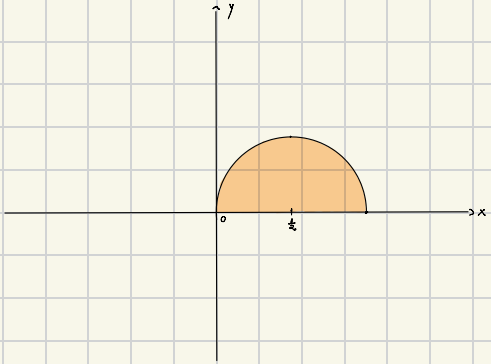
$r \in [0,1]$, $t \in [0, 2\pi)$

$$\iint_D f(x,y)\,dx\,dy = \int_0^{2\pi}\left(\int_0^1 r e^{r^2}\,dr\right)dt = \int_0^{2\pi}\left(\frac{1}{2}e^{r^2}\Big|_0^1\right)dt = \frac{1}{2}(e-1)\cdot 2\pi = \pi(e-1)$$

$$\text{aria}(D) = \iint_D dx\,dy = \int_0^{2\pi}\left(\int_0^1 r\,dr\right)dt = 2\pi\cdot\frac{1}{2} = \pi$$

15. $D = \{(x,y) \in \mathbb{R}^2 \mid x^2 + y^2 \le x,\ y \ge 0\}$, $f(x,y) = xy$. Calculați și aria (D)

$x^2 + y^2 \le x \iff x^2 - x + \frac{1}{4} + y^2 \le \frac{1}{4} \iff \left(x - \frac{1}{2}\right)^2 + (y-0)^2 \le \left(\frac{1}{2}\right)^2 \Rightarrow$ centru $\left(\frac{1}{2}, 0\right)$ și raza $R = \frac{1}{2}$



$x = r\cos t + \frac{1}{2}$

$y = r \sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

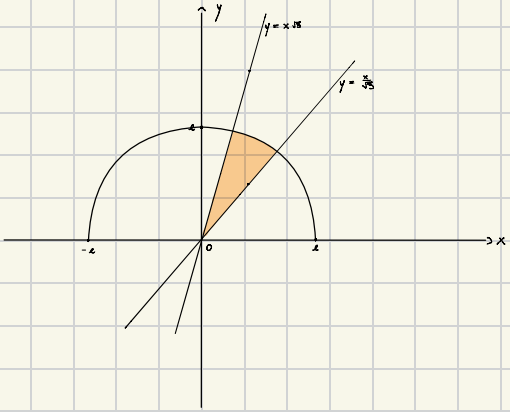
$r \in \left[0, \frac{1}{2}\right]$, $t \in [0, \pi]$

$$\iint_D f(x,y)\,dx\,dy = \int_0^{\pi}\left(\int_0^{\frac{1}{2}} (r^2 \sin t \cos t + \frac{1}{2} r \sin t)\cdot r\,dr\right)dt = \frac{1}{64}\int_0^{\pi} \sin t \cos t\,dt + \frac{1}{48}\int_0^{\pi} \sin t\,dt =$$

$$= \frac{1}{64}\int_0^{\pi} \sin 2t\,dt - \frac{1}{48}\cos t\Big|_0^{\pi} = \underbrace{\frac{1}{256}\cos 2t\Big|_0^{\pi}}_{=0} + \frac{1}{48} + \frac{1}{48} = \frac{1}{24}$$

$$\text{aria}(D) = \frac{1}{2}\pi R^2 = \frac{1}{2}\pi \cdot \frac{1}{4} = \frac{\pi}{8}$$

16. D este domeniul mărginit de cercul $x^2 + y^2 = e^2$, $y = x\sqrt{3}$, $x = y\sqrt{3}$, $x \ge 0$, $f(x,y) = \ln(1 + x^2 + y^2)$



$x = r\cos t$

$y = r \sin t$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$r \in [0, e]$, $t \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

$$\iint_D f(x,y)\,dx\,dy = \int_{\pi/6}^{\pi/3}\left(\int_0^{e} r \ln(1+r^2)\,dr\right)dt = \int_{\pi/6}^{\pi/3}\left(\frac{1}{2}(1+e^2)\ln(1+e^2) - \frac{e^2}{2}\right)dt = \frac{\pi}{12}\left((1+e^2)\ln(1+e^2) - e^2\right)$$

$$\int_0^e x \ln(1+x^2)\,dx = \frac{1}{2}\int_0^e (1+x^2)'\ln(1+x^2)\,dx = \frac{1}{2}(1+x^2)\ln(1+x^2)\Big|_0^e - \frac{1}{2}\int_0^e 2x\,dx = \frac{1}{2}(1+e^2)\ln(1+e^2) - \frac{e^2}{2}$$

17. $\Delta = \{(x,y) \in \mathbb{R}^2 \mid x \geq 0,\ y \geq 0\}$, $f(x,y) = e^{-2(x^2+y^2)}$

$x = r\cos t$

$y = r\sin t$

$r \in [0, \infty)$, $t \in \left[0, \frac{\pi}{2}\right]$

$J(r,t) = r \Rightarrow dx\,dy = r\,dr\,dt$

$$\iint_\Delta f(x,y)\,dx\,dy = \int_0^{\pi/2}\left(\int_0^\infty r\,e^{-2r^2}\,dr\right)dt = \int_0^{\pi/2}\left(-\frac{1}{4}e^{-2r^2}\Big|_0^\infty\right)dt = \frac{1}{4}\cdot\frac{\pi}{2} = \frac{\pi}{8}$$

18. Calculați aria (Δ), unde Δ este domeniul situat în interiorul curbei de ecuație: $(x-2y+3)^2 + (3x+4y-1)^2 = 100$

$$\left.\begin{array}{l} x - 2y = u \\ 3x + 4y = v \end{array}\right| \Rightarrow (u+3)^2 + (v-1)^2 = 100$$

$\Delta^* = \{(u,v) \in \mathbb{R}^2 \mid (u+3)^2 + (v-1)^2 \leq 100\} \Rightarrow \Delta^*$ = interiorul cercului de centru $(-3,1)$ și rază $R = 10$

$x = \frac{2}{5}u + \frac{1}{5}v$

$y = -\frac{3}{10}u + \frac{1}{10}v$

$$J^*(u,v) = \frac{D(x,y)}{D(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{3}{10} & \frac{1}{10} \end{vmatrix} = \frac{1}{50}\begin{vmatrix} 2 & 1 \\ -3 & 1 \end{vmatrix} = \frac{1}{10}$$

$dx\,dy = |J^*(u,v)|\,du\,dv = \frac{1}{10}\,du\,dv$

$$\text{aria}(\Delta) = \iint_\Delta dx\,dy = \frac{1}{10}\iint_{\Delta^*} du\,dv$$

$u = r\cos t - 3$

$v = r\sin t + 1$

$J(r,t) = r \Rightarrow du\,dv = r\,dr\,dt$

$r \in [0, 10]$

$t \in [0, 2\pi)$

$$\int_0^{2\pi}\left(\int_0^{10} r\,dr\right)dt = 100\pi \Rightarrow \text{aria}(\Delta) = 10\pi$$